Week 3 Interim Submission

Healthcare Workflows & Systems Thinking

Project Title: AI-Assisted Early Risk Stratification Emergency Department Triage Support Using Routinely Collected Triage Information

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1. Purpose of Workflow Mapping

The purpose of this Week 3 interim submission is to map the current emergency department triage workflow and identify where an AI-assisted triage support system could realistically fit. This stage is important because a model that performs well in isolation may still fail if it does not match the clinical workflow, staffing realities, data-entry process, and delays of the emergency department.

For this project, the AI system is treated as one component within a larger clinical system. The workflow therefore considers inputs, outputs, actors, constraints and feedback loops. The aim is not only to ask whether an AI model can identify higher-risk patients, but whether it can do so in a way that reduces friction for clinical staff and supports safer decision-making in the actual ED environment.

The workflow is informed by De Freitas et al., who used qualitative observational process mapping to study patient flow in a Caribbean emergency department. Their study is especially relevant because it highlights that ED flow is shaped by work processes, layout, staffing, delays, resources and external departments, rather than by clinical decisions alone [1]. This is important for Mercer General because an AI triage support system must be designed for the workflow that exists, not an idealised version of emergency care.

2. Draft ED Triage Workflow Notes

The ED triage workflow begins when a patient arrives in the emergency department. The patient may arrive by ambulance, walk in independently, be accompanied by a relative or caregiver, or be referred from another health facility. At arrival, staff must first determine whether the patient is critically unwell and needs immediate resuscitation or whether the patient can proceed through the standard registration and triage pathway.

If the patient appears critically unwell, the patient should be directed immediately to the resuscitation room for urgent clinician review, stabilisation and investigations. This route must remain separate from any routine AI-assisted scoring process because immediate clinical action should not be delayed by digital intake, registration or model output.

If the patient is not critically unwell, the next decision is whether the emergency department is the appropriate place for care. Some patients may be redirected to a local health centre if their condition does not require ED-level care. Patients who remain in the ED proceed to registration or intake, where demographic details and the presenting complaint are captured. This is a key early point where information enters the system, but it is also a potential delay point if registration and triage are physically or operationally separated.

After registration, the patient waits for the triage nurse. The triage nurse then assesses the patient and records routine triage information such as presenting complaint, pulse, blood pressure, respiratory rate, temperature, oxygen saturation, GCS and other relevant observations. This is the main point where the proposed AI-assisted system could use routinely collected triage data to support earlier identification of higher-risk patients.

However, ED workflows may include delays and workarounds, including information being recorded before it becomes available in the formal electronic system[1]. This means an AI plug-in cannot assume that all information is immediately available. If vital signs or presenting complaints are missing, delayed or inconsistently entered, the system should recognise that limitation rather than produce a misleading risk flag.

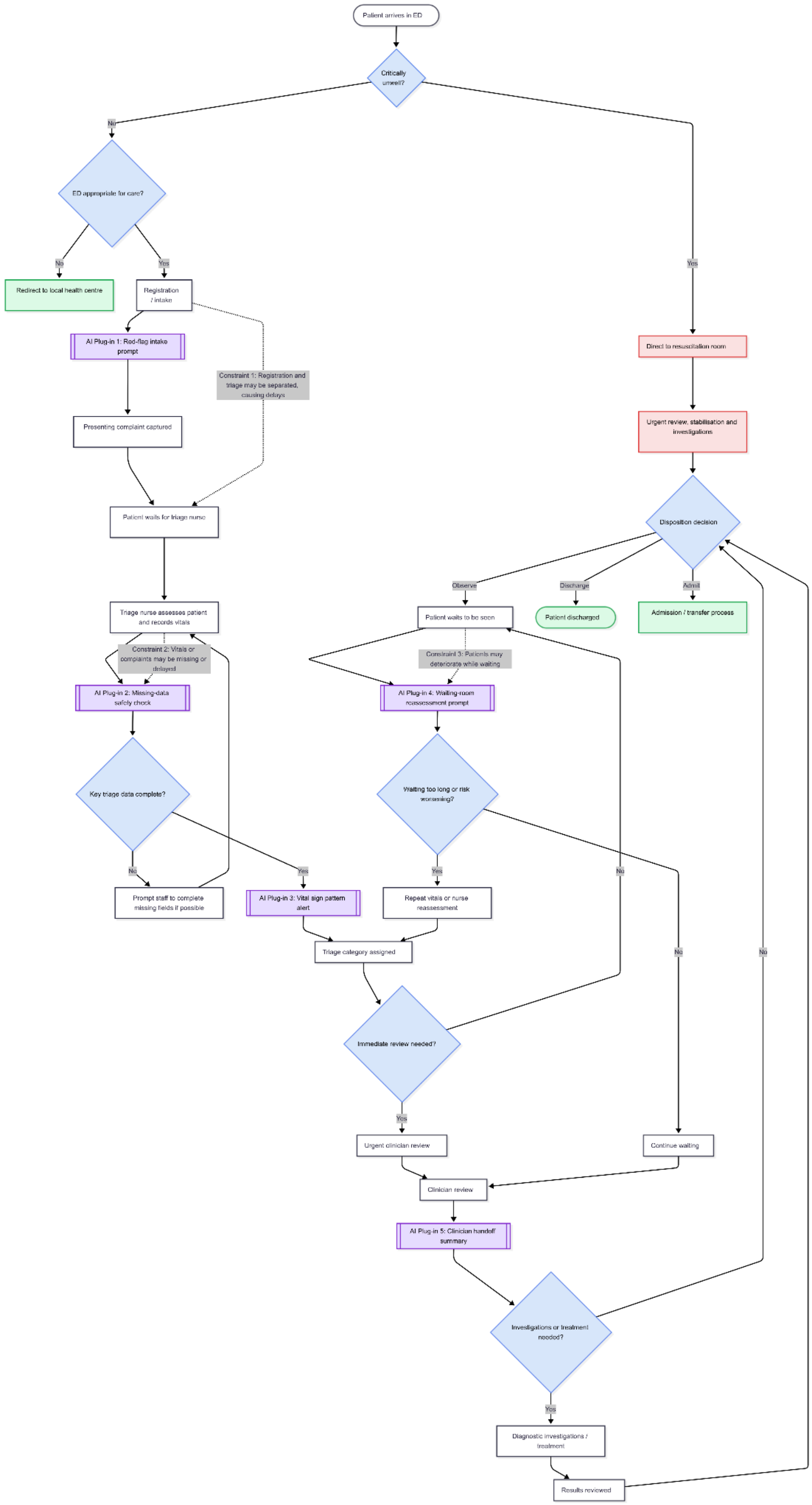
Once triage data is available, the triage nurse or doctor assigns a triage category. At this stage, the AI system could support the clinician by flagging concerning patterns and explaining why the patient may need faster review. The tool should not replace the triage nurse's judgement or assign final priority independently. Instead, it should assist the nurse by making risk information easier to interpret.

After triage, the patient may receive urgent clinician review, wait to be seen, undergo diagnostic investigations or treatment, be observed, discharged, admitted or transferred. Waiting is a major part of the ED process. A patient may wait after registration, after triage, before clinician review, while awaiting investigations, while awaiting results or while awaiting admission or transfer. Because risk can change during waiting, a one-time triage flag may not be enough. A useful system should help identify when reassessment or repeat vital signs may be needed.

3. Initial Workflow Diagram

The initial workflow diagram was created using Mermaid. The diagram shows the patient journey from ED arrival through critical assessment, registration, triage, waiting, clinician review, investigations or treatment and disposition. It also includes five AI plug-in points and three candidate workflow constraints.

Figure 1. Initial ED triage workflow diagram with AI plug-in points and workflow constraints.



Patient arrives in ED
Critically unwell?
No
Yes
ED appropriate for care?
No
Yes
Redirect to local health centre
Registration / intake
AI Plug-in 1: Red-flag intake prompt
Constraint 1: Registration and triage may be separated, causing delays
Presenting complaint captured
Patient waits for triage nurse
Triage nurse assesses patient and records vitals
Constraint 2: Vitals or complaints may be missing or delayed
AI Plug-in 2: Missing-data safety check
Key triage data complete?
No
Yes
Prompt staff to complete missing fields if possible
AI Plug-in 3: Vital sign pattern alert
Triage category assigned
Immediate review needed?
Yes
No
Urgent clinician review
Direct to resuscitation room
Urgent review, stabilisation and investigations
Disposition decision
Observe
Discharge
Admit
Patient waits to be seen
Patient discharged
Admission / transfer process
Constraint 3: Patients may deteriorate while waiting
AI Plug-in 4: Waiting-room reassessment prompt
Waiting too long or risk worsening?
Yes
No
Repeat vitals or nurse reassessment
Continue waiting
Clinician review
AI Plug-in 5: Clinician handoff summary
Investigations or treatment needed?
Yes
No
Diagnostic investigations / treatment
Results reviewed

The diagram is intended to remain readable for a clinical reviewer. It therefore simplifies the full patient flow process into the core steps most relevant to the proposed AI-assisted triage support system. The Mermaid source file and exported diagram image will be committed to the GitHub repository so that the diagram is both readable and version-controlled.

4. Initial AI Plug-In Points

The AI plug-in points were selected based on where information enters the ED workflow, where decisions are made, where delays occur and where risk information may be lost. For each plug-in point, the key question is: what goes in, what comes out and what does the human do next?

AI Plug-In 1: Red-Flag Intake Prompt

Workflow location: Registration or intake
Input: Presenting complaint and obvious red-flag symptoms
Output: Prompt for priority nurse review if concerning symptoms are reported
Human action: Registration or intake staff notify the triage nurse without making a clinical decision

This plug-in is intended to support early recognition of symptoms such as chest pain, shortness of breath, fainting, confusion, severe bleeding or visible distress. It does not allow non-clinical staff to triage the patient. Instead, it supports the handoff from intake to the triage nurse. This is relevant because machine learning and natural language processing methods can support triage performance when presenting complaint information and other triage data are used appropriately [7].

Upstream: Patient has arrived and is being registered.

Downstream: Triage nurse is prompted to review the patient earlier if red-flag symptoms are present.

AI Plug-In 2: Missing-Data Safety Check

Workflow location: After triage data entry
Input: Routine triage fields such as presenting complaint, pulse, blood pressure, respiratory rate, temperature, oxygen saturation and GCS
Output: Prompt showing which key fields are missing or delayed
Human action: Staff complete missing fields where clinically possible or proceed with awareness that the risk assessment is incomplete

This plug-in is included because AI systems are only as useful as the data available at the time they are used. If important fields are missing, the system should not produce a confident risk flag without warning the clinician. This addresses the common workflow problem of delayed, incomplete or inconsistent data entry. Poor workflow fit and additional documentation burden are known barriers to clinical decision support adoption [2,3].

Upstream: The triage nurse has assessed the patient and entered available information.
Downstream: The system either confirms sufficient data for risk support or prompts staff to complete missing fields.

AI Plug-In 3: Vital Sign Pattern Alert

Workflow location: After key triage data are complete
Input: Vital signs and presenting complaint
Output: Plain risk alert identifying concerning clinical patterns
Human action: Triage nurse compares the alert with their clinical judgement and decides whether to escalate, monitor or continue routine triage

This plug-in supports the core project aim: earlier identification of higher-risk ED patients using routinely collected triage information. The tool should avoid presenting only a raw score. Instead, it should explain the reason for concern, such as low oxygen saturation, high respiratory rate, abnormal blood pressure or reduced GCS. Clinician trust in AI depends not only on accuracy, but also on whether the output is understandable and usable in clinical work[4].

Upstream: Triage data have been captured and checked for completeness.
Downstream: The triage nurse uses the alert as decision support before assigning or confirming triage priority.

AI Plug-In 4: Waiting-Room Reassessment Prompt

Workflow location: Patient waiting after triage
Input: Time since triage, risk flag status and whether repeat observations have been recorded
Output: Prompt for repeat vital signs or nurse reassessment if a higher-risk patient waits too long
Human action: Nurse reassesses the patient or repeats vital signs if clinically appropriate

This plug-in is included because ED risk is not static. A patient may appear stable at triage but deteriorate while waiting. Patient waiting and flow delays are prominent features of ED patient journeys, especially when staffing, bed availability or investigations create downstream bottlenecks [1]. A reassessment prompt supports the workflow without assuming that the initial triage decision remains valid throughout the ED stay.

Upstream: Patient has been triaged and is waiting to be seen.
Downstream: If risk worsens or waiting time becomes concerning, repeat observations or reassessment are triggered.

AI Plug-In 5: Clinician Handoff Summary

Workflow location: Before or during clinician review
Input: Presenting complaint, relevant vital signs, risk flag and reason for flag
Output: Short plain-language summary of why the patient was flagged
Human action: Clinician reviews the summary alongside the patient record and clinical assessment

This plug-in is designed to reduce information loss between triage and clinician review. Handoffs are high-risk points because information can be delayed, simplified or lost. A short handoff summary can preserve the reason for an earlier risk flag and help clinicians understand why the patient may need attention. This supports explainability and clinical defensibility while keeping the clinician responsible for final decisions [4,5].

Upstream: Triage assessment and risk flag have already occurred.
Downstream: Clinician receives a concise explanation at review, rather than only a score or alert.

5. Candidate Workflow Constraints

Constraint 1: Registration and Triage May Be Separated, Causing Delays

The workflow may include a gap between registration and triage. During this gap, important information such as presenting complaints or visible distress may be captured informally before the patient is formally assessed by a triage nurse. De Freitas et al. identified layout, work processes and staffing as important factors influencing ED flow in a Caribbean setting [1].

Design implication:
The AI system should not assume that complete clinical data are available at arrival. Early plug-ins should use only data realistically available at that point, such as presenting complaints or red-flag symptoms, and should route concerns to the triage nurse rather than make clinical decisions independently.

Constraint 2: Vital Signs or Presenting Complaints May Be Missing, Delayed or Inconsistently Entered

Triage data may not be immediately available in a complete digital format. Some information may be captured verbally or on paper before it is entered into an electronic system. If an AI tool depends on complete real-time data, it may fail or produce unreliable output when fields are missing.

Design implication:

The system should include a missing-data safety check before producing a risk flag. It should clearly show when the risk assessment is limited by missing information. This reduces the risk of false reassurance and supports safer use of AI in a real clinical environment.

Constraint 3: Patients May Deteriorate While Waiting

The ED workflow includes multiple waiting points: waiting for triage, waiting for clinician review, waiting for investigations, waiting for results and waiting for admission or transfer. If the AI tool only produces one static alert at triage, it may miss deterioration that occurs later in the patient journey.

Design implication:
The system should support reassessment prompts for higher-risk patients who wait too long or have not had repeat observations. This allows the AI tool to support patient flow and safety beyond the first triage decision.

Constraint 4: Alert Fatigue and Poor Trust May Lead Clinicians to Ignore the System

If the system produces too many alerts, unclear alerts or alerts that do not fit the workflow, clinicians may ignore or override them. Alert fatigue is a recognised issue in clinical decision support systems, and inappropriate alerts can interrupt clinical work rather than improve it [6].

Design implication:
The system should prioritise high-value alerts, explain why the patient was flagged and avoid creating unnecessary steps for triage nurses. The tool should reduce cognitive load at a bottleneck rather than add work.

6. Initial Stakeholder Considerations

Although the final proposal will include a more developed stakeholder map, the following stakeholders have already been identified because each may accept, reject or reshape the proposed system.

Stake holder	Main Concern	What Success Looks Like
Triage nurse	The system must support speed and accuracy without adding extra steps during a busy shift.	The nurse sees a clear, useful alert that supports their judgement and does not slow triage.
ED consultant or doctor	The system must provide clinically defensible reasoning, not unexplained scores.	The clinician can understand why a patient was flagged and decide whether escalation is justified.
Registration or intake staff	The system must not require non-clinical staff to make clinical decisions.	Staff can enter presenting complaint information and red-flag symptoms without being asked to triage.
Porter or support staff	The system should not create extra movement, scanning or documentation tasks.	The tool does not depend on support staff completing unfamiliar digital steps before patient care moves forward.
Patient or patient advocate	The system must be fair, transparent and not delay care for patients who present calmly but are clinically at risk.	Patients benefit from earlier recognition of risk without being disadvantaged by missing data, language barriers or biased assumptions.
Clinical IT or EHR lead	The system must be secure, auditable, reliable and able to fit existing documentation processes.	The tool can be reviewed, monitored and safely integrated without exposing private data or creating unreliable workflow dependencies.

This stakeholder mapping reflects the idea that technical performance alone is not enough. A clinically useful AI tool must fit the needs of the people who act on it. If the triage nurse ignores the alert, the tool has no clinical value even if the model performs well in retrospective testing.

7. Initial Friction Points

The following friction points were identified from the draft workflow diagram and workflow notes:

1. **Data may not exist at the moment the AI is expected to fire.**
   For example, a risk model cannot use complete vital signs if vital signs have not yet been taken or entered.

2. **The triage nurse may not have time to interact with a separate tool.**
   A tool that requires extra screens, forms or manual confirmation may fail during high-pressure periods.

3. **Alerts may become noise if they are too frequent or poorly explained.**
   This could reduce trust and cause clinicians to ignore the system.

4. **Handoff points may lose important information.**
   A patient flagged during triage may later be reviewed by another clinician who does not understand why the patient was flagged.

5. **Waiting time changes risk.**
   A patient's status may worsen while waiting, so one-time triage scoring may not be sufficient.

6. **Local workflow realities may differ from international AI literature.**
   ED flow in Caribbean settings may involve different staffing, layout, infrastructure, patient mix and resource constraints from the settings where many AI tools are developed or tested [1]

References

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